

Exhibit CC-131

SAMSUNG

Product brief

Efficiently powering mobile innovation

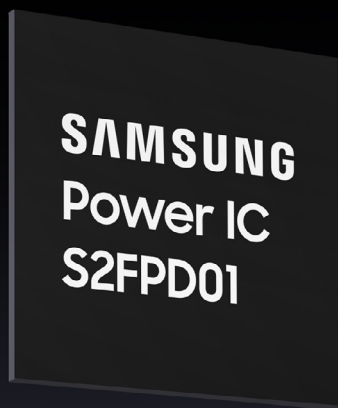
S2FPD01 Power IC

Highlights

Highly efficient architecture to reduce power consumption

Fast transients for output voltage accuracy

Current capability with the best performance for fast data rate and high density DIMMs



Optimized for servers and data centers

The S2FPD01 offers optimal performance for data centers and enterprise servers that run analytics, machine learning, and other computing tasks. The new PMIC for DDR5 module is equipped with a high-efficiency hybrid gate driver for improved power efficiency and load-transient responses.

Low power consumption and fast transient response

By implementing a high-efficiency hybrid gate driver, the ICs can offer up to 91% power-efficiency. This scheme allows the DC voltage to decrease from high to low with a fast transient response to changes in the output load current, and adapts the conversion accordingly to efficiently regulate its output voltage at near-constant levels. The control scheme also features pulse width modulation and pulse frequency modulation to minimize delays and malfunctions during mode switching.

Optimized current capability

Each IC has 4 bucks (SWA/SWB/SWC/SWD) and 2 LDOs. The current capability of each buck is 3A. Dual phase mode (SWA+SWB) supports 6A. It is optimized for fast data rate and high density DIMMs for cost effectiveness.

JEDEC-compliant PMIC

The S2FPD01 fully supports JEDEC standard, as well as PMIC50x0 (Power Management IC Specification Rev1.8, JESD301-1.)

Specifications

	S2FPD01
VIN_BULK (VINA/INB/VINC/VIND)	4.25V ~ 15V
VIN_MGMT	3.0V ~ 3.6V
Power Management	Input UVLO/OVLO, Thermal Shut Down function Read/Write I2C/I3C for Register Control Interface up to 1MHz for I2C and 12.5MHz for I3C Error Log Count & Storage Built-in Nonvolatile Memory registers Dynamic Voltage Scaling controlled by I2C/I3C Programmable Soft Start & Soft Stop Time Current Limit, Over/Under voltage protection Power Meter
Current Capability	3A per each phase
SWA+SWB	Programmable output with 5mV step (0.6V ~ 1.435V) Switching Frequency : 0.5MHz to 1.25MHz
SWC	Programmable output with 5mV step (0.6V ~ 1.435V) Switching Frequency : 0.5MHz to 1.25MHz
SWD	Programmable output with 5mV step (1.5V ~ 2.835V) Switching Frequency : 0.5MHz to 1.25MHz
LDO1	Programmable output voltage 1.7/1.8/1.9V Output current capacity of 25mA
LDO2	Programmable output voltage 0.9/1.0/1.1/1.2V Output current capacity of 20mA

* All product specifications reflect internal test results and are subject to variations by user's system configuration. Actual performance may vary depending on use conditions and environment.
* All product images shown are for illustration purposes only and may not be an exact representation of the product.
* Samsung reserves the right to change product images and specifications at any time without notice.

Copyright © 2022 Samsung Electronics Co., Ltd. All rights reserved. Samsung is a registered trademark of Samsung Electronics Co., Ltd. Specifications and designs are subject to change without notice. Nonmetric weights and measurements are approximate. All data were deemed correct at time of creation. Samsung is not liable for errors or omissions. All brand, product, service names and logos are trademarks and/or registered trademarks of their respective owners and are hereby recognized and acknowledged.

About Samsung Electronics Co., Ltd.

Samsung Electronics Co. Ltd inspires the world and shapes the future with transformative ideas and technologies. The company is redefining the worlds of TVs, smartphones, wearable devices, tablets, digital appliances, network systems, memory, system LSI and LED solution. For more information about the Samsung Semiconductor products, visit semiconductor.samsung.com.

